

NISHANTH ANANTHA

Technical Program Manager · Physical AI & Cloud Platform · Stateful Systems · Agile Delivery

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PROFESSIONAL SUMMARY

Staff TPM with 14 years owning complex, cross-functional technical programs end-to-end — across cloud API platforms, mobile/web infrastructure, AI/ML systems, and distributed software environments. Deep Agile and waterfall expertise (PMI-PMP, SAFe): ran engineering planning, standups, retrospectives, and sprint ceremonies while keeping JIRA sprint state meticulously reflective of ground truth. Proven at identifying project risks, resolving critical dependencies, and coordinating across product, engineering, and QA to deliver high-quality software on time. Built execution frameworks from scratch in ambiguous, high-growth, startup-like environments — lifting release success from 29% to 94% across 30+ programs. Strong AI fluency: shipped production agentic AI system (Claude API + MCP) and RAG conversational agent (82% CSAT). Certified: PMI-PMP, Anthropic Claude API, MLOps (DeepLearning.AI), AWS, IBM GenAI & LLMs.

KEY IMPACT

- **Full development lifecycle ownership:** owned cloud API, web, and mobile software programs end-to-end — requirements through release — coordinating product, engineering, QA, and operations across tightly coupled workstreams.
- **Stateful platform & dependency coordination:** managed critical path across 10+ product teams on 100+ field machine programs; regularly aligned contributors on project risks, dependency schedules, and JIRA sprint accuracy.
- **94% release success from 29%:** built Go/NoGo phase gate framework, cross-functional readiness checklists, and release cadences from scratch in 18 months with no prior process — became organizational standard across HW, FW, and SW.
- **Agile + waterfall expert:** ran PI planning, sprint planning, standups, retrospectives, and CCB ceremonies across SAFe multi-team environments; JIRA sprint health and burn-down maintained as trusted signal for Director/VP stakeholders.
- **AI-forward execution:** used GenAI tools daily for program artifact drafting, risk synthesis, and status reporting; shipped production agentic AI workflow (Claude API + MCP) and RAG agent (82% CSAT, 65% deflection) in live environments.
- **Engineering team leadership:** built and led cross-functional teams of 7–20 across US, Europe, and Asia; moved teams from reactive waterfall to structured SAFe Agile; mentored junior engineers and TPMs.
- **M+ customer savings + M+ infrastructure ROI:** rapid cross-functional escalation resolution; data-backed investment decisions communicated to Director/VP with clear risk trade-offs and ROI framing.

PROFESSIONAL EXPERIENCE

Staff Technical Program Manager — Cloud Platform & AI/ML Programs | ASML, San Diego, CA Jan 2024 – Present

Team: 20 · Stateful platform delivery, AI/ML tooling, cloud API and software release programs

- Owned overall development lifecycle of complex stateful platform programs — cloud API, web, and mobile initiatives — coordinating product, engineering, and QA to ensure well-run software development, test, and release processes.
- Regularly aligned contributors on project risks and critical dependencies — ran engineering planning, standups, and retrospectives; kept JIRA sprint state reflective of actual work across 20+ concurrent workstreams.
- Worked across teams to build and maintain comprehensive dependency schedules — identified logical gaps, coordinated resolution, and escalated blockers to ensure timely, high-quality delivery.
- Partnered with product and senior engineers to develop strategy and roadmap for compelling platform capabilities — framed technical design tradeoffs, provided data-informed input to product planning cycles.
- Built AI-powered program health system (Claude API + MCP + ensemble ML) — synthesizes 13 live signals (GitHub CI, JIRA sprint state, Smartsheet program health) into Go/NoGo risk scores with SHAP explainability; adopted by Director/VP as standard release infrastructure.
- Shipped production RAG Conversational AI Agent with full eval pipeline — 82% CSAT, 65% user deflection, 1.1s p50 latency; designed user feedback loop from day one for continuous optimization.
- Managed M+ validation infrastructure program — 70% field defect reduction; ROI and risk tradeoffs communicated clearly to Director and VP stakeholders.
- Led and mentored team of 20 — established operating cadences, facilitated leadership syncs, synthesized discussions into action items with clear owners and accountability.

Sr. Technical Program Manager — SW/HW Platform & Cloud Release Programs | ASML, San Diego, CA Mar 2019 – Dec 2023

Team: 10 · Full software/hardware integration and release lifecycle across 100+ field machines globally

- Owned software development lifecycle across 100+ field machines — coordinated alpha/beta campaigns, enforced quality gates, managed critical path across 10+ product teams; delivered fully tested products to customers on time.

- Built full program execution framework from scratch — release success from 29% to 94% in 18 months across 30+ programs; Go/NoGo operating model, quality gates, and phase gate criteria became organizational standard.
- Ran Agile ceremonies across multiple teams — sprint planning, standups, retrospectives, PI planning (SAFe); owned product backlog and JIRA sprint hygiene as a trusted signal for engineering leadership.
- Prioritized and managed multiple concurrent projects — maintained schedule discipline, surfaced risks proactively, and resolved interdependencies before they became blockers.
- Drove CI/CD transformation org-wide (Jenkins, GitHub Actions, Docker, TF Serving) — 65% faster software delivery; standardized build configuration and automated verification gates across all programs.
- Reduced customer-facing defects 50%; M+ escalation resolution — structured root-cause analysis, corrective actions enforced at Change Control Board; chaired CCB across US, Europe, and Asia.

Sr. Software Engineer / Team Lead / Scrum Master | ASML, San Diego, CA Nov 2016 – Mar 2019

Team: 7 · Engineer-to-TPM arc; Python automation and SAFe Agile transformation

- Shipped Python automation to production while owning SAFe Agile planning, backlog management, and release governance — lifted release success from 29% to 94% in 18 months; saved M+ through rapid escalation resolution.
- Moved team from waterfall to SAFe Agile — built sprint discipline, JIRA hygiene, and cross-functional coordination cadences from scratch; awarded for Leadership and Ownership.

Sr. Platform Test / Embedded SW Engineer | Caterpillar · Volvo Group · Abbott · SDP Engineering 2009 – 2016

- Python/C automation, HIL testbenches, CAN protocols, FPGA-based real-time systems across automotive, industrial, and medical domains; 00K+ escalation resolution; Volvo Patent Idea of the Year — Global Patent Council (2012).

AI SYSTEMS BUILT & DEPLOYED

- **Agentic Workflow System (Production):** Claude API + MCP + GitHub REST + JIRA + Smartsheet + PyTorch + GBM + DistilBERT + SHAP + FastAPI. Agentic loop routes tool calls across live cloud data sources; synthesizes 13 signals into Go/NoGo risk score. Safety hardened: Pydantic validation, token bucket rate limiting, DDoS prevention.
- **RAG Conversational AI Agent (Production):** HuggingFace + ChromaDB + TF Serving + Streaming Inference + W&B LLM Evals. Full eval arc: BLEU/ROUGE offline → 82% CSAT, 65% deflection, 1.1s p50 latency online.

SKILLS & TOOLS

Platform & Cloud Program Management: Cloud API, mobile, web platform delivery; stateful system lifecycle; distributed systems; dependency tracking; phase gate criteria; Go/NoGo frameworks; risk management

Agile & Process Expertise: SAFe, Agile, waterfall, PI planning, sprint ceremonies, JIRA administration, backlog management, CCB; PMI-PMP (Active); executive communication (Director/VP)

AI & Technology Fluency: Claude API (Anthropic certified), MCP, agentic systems, RAG, PyTorch, HuggingFace, TF Serving, MLOps (DeepLearning.AI), IBM GenAI & LLMs, AWS (Certified), Docker, GitHub Actions, CI/CD

Languages & Tools: Python, SQL, JIRA, Smartsheet, MS Project, Confluence, FastAPI, REST, Power BI, Docker

EDUCATION

M.S. Electrical & Electronics Engineering — University of Southern California (USC) | May 2014

B.E. Electronics & Communication Engineering — SJCE, VTU, India | Jul 2009

CERTIFICATIONS

Machine Learning in Production (MLOps) — DeepLearning.AI (Jun 2026) · Building with Claude API — Anthropic (Apr 2026) · Claude on Google Cloud Vertex AI — Anthropic (May 2026)

W&B LLM Evals (May 2026) · IBM Generative AI & LLMs (Feb 2026) · Deep Learning with TensorFlow — Kaggle (Apr 2026) · PMI-PMP (Active) · AWS Certified Cloud Practitioner (2024)